

AlarmCare: Exploring Mobile Alarm as a Supportive Communication Channel Between Peers

Seongsoo Kim
Chungnam National University
Daejeon, Republic of Korea
seongsoo.kim025@gmail.com

Byeongju Kim
Chungnam National University
Daejeon, Republic of Korea
catchwall3203@naver.com

Yuju Kang
Chungnam National University
Daejeon, Republic of Korea
yuju@o.cnu.ac.kr

Minsam Ko
Hanyang University ERICA
Ansan, Republic of Korea
minsam@hanyang.ac.kr

Jaejeung Kim
Chungnam National University
Daejeon, Republic of Korea
jjkim@cnu.ac.kr

ABSTRACT

The use of ubiquitous computing for supportive communication in the constantly evolving field of mobile technology to improve mental health support between peers is significant. We designed and implemented AlarmCare, a mobile application for creating alarms on peer's smartphone with a mentally supporting text and voice message. We conducted a one-week, explorative study with eighteen participants to understand how mobile alarms were used as a supportive communication channel between peers. The findings indicate that peer created alarms decreased depressive emotions and fostered positive socially supporting emotions including but not limited to 'Sense of being cared for' and 'Sense of togetherness'. Our study reveals the potential of mobile alarms as a viable tool which can be utilized in daily communications for enhancing mental health.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Ubiquitous and mobile devices**.

KEYWORDS

Peer-created Alarm, Supportive Communication, Mental Health, Voice Message

ACM Reference Format:

Seongsoo Kim, Byeongju Kim, Yuju Kang, Minsam Ko, and Jaejeung Kim. 2024. AlarmCare: Exploring Mobile Alarm as a Supportive Communication Channel Between Peers. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3677594>

1 INTRODUCTION

Depression is a common mental disorder. According to the World Health Organization, approximately 280 million people in the world is suffering from depression[18]. However, as much as 39% of adults

with depressive symptoms did not actively seek professionals for treatment in the U.S.[15]. Numerous factors influence the willingness to receive treatment, including attitudinal and structural barriers. Mobile sensing for predicting changes in the depressive of mood, and offering just-in-time stress intervention with mobile and wearable sensing with the ubiquitous computing technology is very promising [2, 9]. Such approach is a promising cost-effective and accessible way of interventions. However, some argue that self-help apps are not effective in the real-world settings[28]. There are supporting tools that enable people to receive mental health care from others remotely. These tools utilize various techniques such as text messaging, voice, and video transmission through mobile devices and PCs[3, 26]. However, most of existing research primarily focuses on mental health support between clinicians and patients[4, 23, 30]. Unlike the professional clinician care, everyday supportive communication channels such as phone calls, text messaging, and social media offer low-cost, easily accessible mental health support[10–13, 22]. These channels have the potential to reduce depressive symptoms and promote mental and physical well-being among peers (friends, lovers, etc.)([11, 14].

In this study, we attempt to explore a new channel for supportive communication by designing AlarmCare, employing peer-created alarms as a daily mental health support between peers. Unlike push notifications from text messaging and social media, which primarily alert users with brief sounds or vibrations, the alarm feature on smartphones offers more persistent music or sound until the user's explicit dismissal. Also, alarms can deliver scheduled messages appropriate for repeated daily care. Due to these characteristics, alarms have been used as self-care tools for sleep, medication, and other purposes[5, 17, 24]. We decided to use the human voice message as a communication method available when the alarm goes off. This was inspired by telephone counseling[6, 8], a traditional mental supportive communication known as a feasible and acceptable alternative to the specialized depression therapy[27]. Accordingly, the users of AlarmCare can use the alarm to send a voice message to their peers at specified times during the day. By conducting an in-the-wild deployment study with AlarmCare, we aim to understand the following:

- What types of peer-caring alarms are created?
- What types of emotions are elicited by the alarms?
- How do they impact users' depressive emotions?

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia.

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3677594>

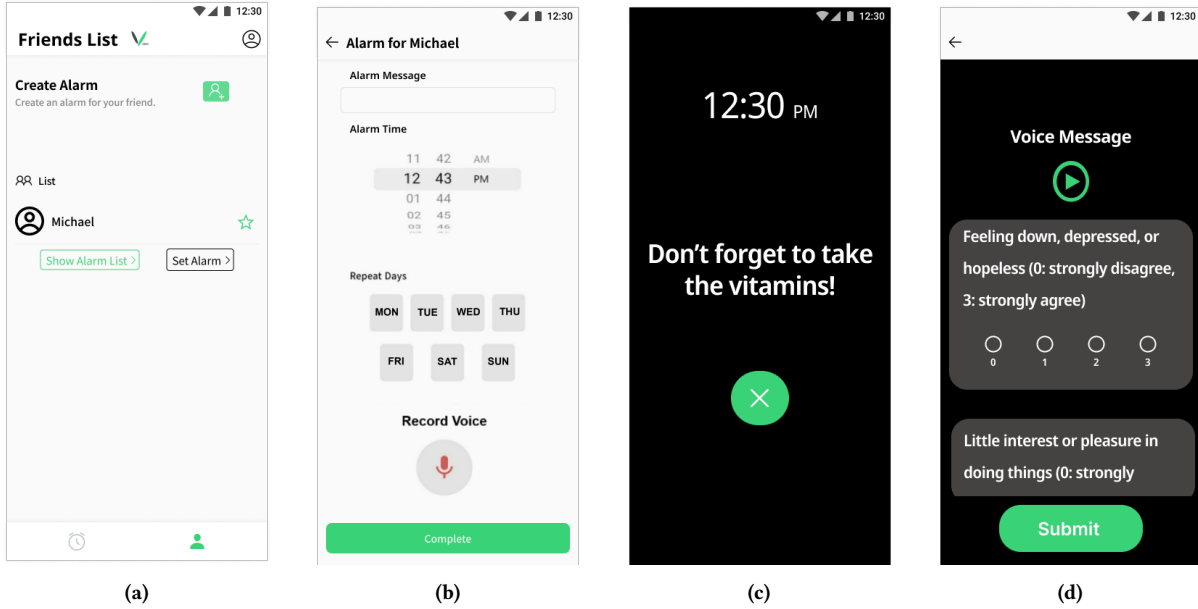


Figure 1: Screenshots of the AlarmCare: (a) Friends List for adding or managing alarms, (b) Alarm setting with voice recording, (c) Alarm triggered screen(text message + default alarm sound), (d) Voice message player and EMA questionnaire for emotion assessment

2 ALARMCARE APP DESIGN

We implemented an Android application "AlarmCare" for experiment control and data collection. AlarmCare was designed to set and trigger alarms on a peer's smartphone with a message and a voice recording. A pair of users registered as peers can set alarms to be triggered on each other's smartphones at a certain time.

The process is as follows. First, the user selects a peer from the list and selects 'Set Alarm' (Figure 1a). Next, the user fills in the alarm message, time, the days of the week to be activated, and records a voice message to be played(Figure 1b). When the alarm goes off, the default alarm sound (or vibration) activates along with the text message (alarm message) the user left for the peer (Figure 1c). Once the peer close the text message, the voice message can be played followed by three EMA questionnaire for emotion assessment(Figure 1d). We did not allow the voice message to be played without the receiver's explicit intention for two reasons: to make sure the receiver actually listens to the message, and to prevent privacy sensitive content being played without permission.

3 METHODS

3.1 Participants

A total of 9 groups of two individuals as a team were recruited via an online community. Inclusion criteria were one of the team member is experiencing from minimal to moderate severe levels of depressive emotions measured by PHQ-9 in the recruitment survey[1]. As a result, 18 participants were selected and classified into 5 minimal, 7 mild, and 4 moderate, 2 moderate severe levels. Among the 9 groups, 7 had a friend relationship, 1 team had

a sister relationship, and 1 team had a romantic relationship (Female=9, Male=9; Mean age=22.06, SD=2.92). All participants were comfortable using smartphones and frequently used alarm apps in day-to-day usage.

3.2 Experimental Procedure

We conducted a 7-day in-the-wild experiment. Participants were informed that AlarmCare usage logs and voice recordings would be collected during the experiment. The researchers used the Android Quick Share feature to install the experimental app, AlarmCare, on the participants' smartphones. Then the paired partners registered each other in AlarmCare. We encouraged the participants to set more than one alarm per day, but this was not mandatory. Prior to the experiment, the participants were conducted the PHQ-9 to screen for depression. Additionally, Ecological Momentary Assessment (EMA)[25] was used capture the user's depressive emotion as well as the perceived social support upon receiving the alarm and listening to the voice message from their peer. At the end of the 7-day experiment, PHQ-9 was once again measured for a pre-post comparison of depressive emotions. A semi-structured interview was conducted for obtaining qualitative data of the overall experience.

The EMA questions assessed participants' depression levels and their sense of being cared for. The survey items included two questions from the PHQ-2, a validated short form of the PHQ-9, and the other question asked, "I felt like I am being cared for." This question aimed to evaluate the degree of perceived social support and sense of community enhanced by supportive communication[16, 19, 29]. The responses from EMA questions were collected for experimental assessment only and were not shared with the peer.

Table 1: Summary of alarm types, categories, and 4-point Likert Scale ratings of Perceived Social Support

Type	Category	0	1	2	3	Sum	Mean (SD)
Routine	Wake up	0	0	6	13	19	2.68 (0.03)
	Bedtime	0	0	5	10	15	2.67 (0.04)
	Meal	0	0	9	6	15	2.40 (0.03)
	Medication	0	2	5	3	10	2.10 (0.20)
	Work out	0	0	5	1	6	2.17 (0.12)
Non-Routine	Sender-centered	0	2	17	16	35	2.40 (0.07)
	Receiver-centered	0	0	3	7	10	2.70 (0.06)

4 RESULTS

4.1 Alarm Types

Routine: During the experiment period, a total of 110 alarm instances were created and triggered. Among the total alarms created, 65 (59.1%) alarms were classified as ‘routine’, which we defined as a repeated daily activities. Specific categories under the routine alarms were analyzed based on the alarm messages and voice recording contents. They were categorized into reminders to wake up, encouragement to eat, suggestions for workouts, encouragement to take medication, and reminders to go to bed. The wake-up category had the most usage records(29.2%), followed by bedtime(23.0%) and meal time(23.0%) reminder. Note that the alarm messages were written as more ‘caring’. For example, “Have a nice meal” or “Take a good rest(sleep)”. Medication(15.4%) and workout(9.2%) suggestions were created to care for peer’s health. The voice messages contained in this type of alarm had an average duration of 2.84 seconds(SD=2.71), which is typically within one sentence length. For instance, the majority of alarm voice messages for wake-up reminders said, “Wake up, time to go to class”.

Non-Routine: A total of 45 (40.9%) belonged to ‘non-routine’ alarms, which was used for one-time communication similar to text messaging. We further categorized them into the ‘sender-centered(77.8%)’ and ‘receiver-centered(22.2%)’ alarms. The duration of the voice messages in this ‘non-routine’ category was 6.03 seconds (SD=8.18). Some examples from sender-centered alarm voice messages were “I’m coming home after work” and “The test I took today was difficult”. Receiver-centered alarm voice messages featured phrases like “Take care when you go home!” and “You must have been through a lot today”.

4.2 Emotion Types

We conducted a thematic analysis on the interview responses on the emotions after receiving the alarm. It was categorized into five types of emotions: (1) Sense of Being Cared for, (2) Sense of Expectation, (3) Sense of Togetherness, (4) Sense of Disturbance, (5) Sense of Cheerfulness (Table 2). We further explored what aspect of the alarm triggered such emotions.

Sense of Being Cared for: This was the most prominent emotion mentioned by the participants. 11 participants mentioned that the actual voice from the peer was emotionally caring. 2 participants said that it was the textual content of the alarm message that was perceived as caring.

The EMA results show that the participants felt that they are being cared for after receiving the alarm ($M=2.47$, $SD=0.02$). It is notable that both routine ($M=2.48$, $SD=0.04$) and non-routine ($M=2.47$, $SD=0.06$) alarms were emotionally supporting. However, there were two negative responses each from Medication and Sender-centered alarm types. The messages of negative cases for medication were a message to take medicine in the evening, and it was difficult to take medicine immediately because the receiver had an appointment then. Also, the negative case from the Sender-oriented alarm was the case of announcement for a receiver that the sender was going somewhere to study for the exam, and when the receiver was doing something urgent, the alarm arrived and was a disturbance.

Sense of Expectation: Five participants (27.8%) expressed that they were eager to know what message is awaiting in the voice message. Note that the participants were able to see the ‘alarm message’ before the alarm is triggered. However, the voice message could only be played when the alarm goes off. P3 said, “*I felt curious and looked forward to what kind of voice recording it would be*”. P13 said, “*I wanted to listen to the voice message even before it rang because I was curious about what the other person had recorded*”.

Sense of Togetherness: Four participants (22.2%) mentioned that the voice message gave them the sense of being together which mentally supported them at the time of depression. P9 stated, “*I usually feel depressed at night. But at that moment when I heard the voice on the alarm, I felt like someone was with me and it comforted my feelings*”.

Sense of Disturbance: Three participants (16.7%) experienced disturbance from the alarms. This was the only negative emotion feedback, which the participants stated that it was a frustrating experience. P12 said specifically, “*I felt disturbed when the alarm went up at an unexpected time*”.

Sense of Cheerfulness: Two participants (11.1%) experienced a sense of cheerfulness when the voice message contained an entertaining music and singing. P3 stated, “*When I heard amusing voice messages—like her singing—I was amused and happy*”.

4.3 Impact on Depression

15 out of 18 participants’ PHQ-9 score decreased. There was a statistically significant difference between PHQ-9 pre-post rating scores. The Shapiro-Wilk test was utilized to verify the normality of the data. The PHQ-9 scores decreased from 7.9 to 3.9 ($t=3.647$, $p<0.01$). Also, the effect size Cohen’s d was 0.92, indicating the use of AlarmCare had a large effect on decreasing depression. Furthermore,

Table 2: The participants’ emotional descriptions after receiving the alarm, the factors that made them feel, the count of participants corresponding to each factor, and the examples

Description of emotion	Emotional elements (Count)	Examples
Sense of Being Cared for	Voice message (11)	<i>“I had the feeling that she was taking care of me when I heard her voice.” (P17)</i>
	Alarm message screen (2)	<i>“The alarm message that peer set created the impression that they were cared about me.” (P9)</i>
Sense of Expectation	Voice message + delayed delivery (5)	<i>“I felt curious and looked forward to what kind of voice recording it would be.” (P3)</i>
Sense of Togetherness	Voice message + just-in-time delivery (4)	<i>“... but at that moment when I heard the voice on the alarm, I felt like someone was with me and it comforted my feelings.” (P9)</i>
Sense of Disturbance	Unpredictability of alarm time (3)	<i>“I felt disturbed when the alarm went off at an unexpected time.” (P12)</i>
Sense of Cheerfulness	Voice message + singing/music (2)	<i>“When I heard amusing voice messages—like her singing—I was amused and happy.” (P3)</i>

the EMA responses to PHQ-2 questionnaire upon examination of the PHQ-2 responses provided by the 18 respondents to the EMA survey, it was discovered that, out of 110 answers, 6 times (5.4%) selected 2 or 3 points (agree - strongly agree) for any one of the two questions. For each of the two questions, 104 responses (94.6%) gave a score of 0 or 1 (strongly disagree - disagree).

5 DISCUSSION

5.1 Advantages of Asynchronous Communications

The asynchronous nature of alarm based communication is similar to the asynchronous telepsychiatry(ATP) techniques have been developed as an alternative to STP[30] which is advantageous in alleviating geographical and scheduling barriers to access[3]. Particularly, the routine alarm types demonstrate the potential of action-facilitating support among the supportive communication methods[20] by encouraging the receiver to listen to the voice message at certain times and take actions that correspond to the routine.

Additionally, the asynchronous feature makes it convenient for both the sender and the receiver of the alarm. P1 mentioned that *“Instant messengers can communicate freely and on various topics, but the alarm itself was focused on communication with a clear focus on time. So I was able to record only the necessary parts and words without recording long voice messages”*. P10 said, *“Compared to the telephone call, I liked that the receiver could hear the peer’s voice without doing anything specific, such as answering the phone”*. P18 mentioned the benefit of alarms over telephone calls by saying, *“The telephone calls need to be made at a certain time of the day for routine check-ups, but the alarm automatically repeats at that time every week so that the message can be heard”*. In conclusion, AlarmCare’s asynchronous features enables the users to listen to peer’s voices without having them time each other or do specified actions, and there’s no need to be concerned about using it for an extended period of time, which can be an issue while synchronous communicating via video[21].

5.2 Voice Message

We observed various forms of voice messages and their impact on eliciting positive emotions. The voice message, unlike text, consists both sound and textual information. Particularly the sound factor has multiple dimensions including the identity of the peer, emotion expression, which are very delicate emotion eliciting elements. Also, the recording is not limited to the ‘voice’, but also non-human voices such as music. Therefore, the success of positive emotion evoking depends on the voice message creator. We argue that providing a guideline or functional support to guide the alarm creator to target a particular emotion the receiver want to experience could contribute to more successful emotion support.

We also uncovered different types of emotions evoked by the alarm delivery timing. The combination of the timing and voice message may elicit various types of positive emotions not limited to the ones we listed in current study. On the other hand, some voice messages, such as medicine and receiver-centered alarms with low perceived social support scores, did not take the receiver’s circumstances into account at the time of the alarm. This implies the use of contextual awareness[7] to send messages appropriate timing could boost the postive experience and effectiveness.

6 CONCLUSION

In this study, we have explored how mobile alarm can be a supportive communication channel between peers. Our preliminary study results reveal the it elicited various types of positive emotions which can reduce depressive emotions. There are much more to uncover in understanding the dynamics of mobile alarm mediated communication. By extending our current work, we envision the mobile alarm, which is one of the most frequently used mobile application, could serve as a more targeted mental health supporting tool to be integrated into the people’s daily lives.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea Grant funded by the Korean Government (NRF-2022S1A5A2A 03050752).

REFERENCES

- [1] Bruce Arroll, Felicity Goodyear-Smith, Susan Crengle, Jane Gunn, Ngaire Kerse, Tana Fishman, Karen Falloon, and Simon. Hatcher. 2010. Validation of PHQ-2 and PHQ-9 to screen for major depression in the primary care population. *Ann Fam Med* 8, 4 (2010), 348–53. <https://doi.org/10.1370/afm.1139>
- [2] Luca Canzian and Mirco Musolesi. 2015. Trajectories of depression: unobtrusive monitoring of depressive states by means of smartphone mobility traces analysis. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Osaka, Japan) (*UbiComp '15*). Association for Computing Machinery, New York, NY, USA, 1293–1304. <https://doi.org/10.1145/2750858.2805845>
- [3] Steven Chan, Luming Li, John Torous, David Gratzner, and Peter M. Yellowlees. 2018. Review of Use of Asynchronous Technologies Incorporated in Mental Health Care. *Current Psychiatry Reports* 20, 10 (2018), 85. <https://doi.org/10.1007/s11920-018-0954-3>
- [4] Patricia V. Chen, Ashley Helm, Steve G. Caloudas, Anthony Ecker, Giselle Day, Julianna Hogan, and Jan Lindsay. 2022. Evidence of Phone vs Video-Conferencing for Mental Health Treatments: A Review of the Literature. *Current Psychiatry Reports* 24, 10 (2022), 529–539. <https://doi.org/10.1007/s11920-022-01359-8>
- [5] Sara Chew, Pauline Siew Mei Lai and Chirk Jenn Ng. 2020. Usability and Utility of a Mobile App to Improve Medication Adherence Among Ambulatory Care Patients in Malaysia: Qualitative Study. *JMIR Mhealth Uhealth* 8, 1 (31 Jan 2020), e15146. <https://doi.org/10.2196/15146>
- [6] Diana Dorstyn, Jane Mathias, Linley Denson and Marie. Robertson. 2012. Effectiveness of telephone counseling in managing psychological outcomes after spinal cord injury: a preliminary study. *Arch Phys Med Rehabil* 93, 11 (2012), 2100–8. <https://doi.org/10.1016/j.apmr.2012.06.002>
- [7] Luis Claudio Gubert, Cristiano André da Costa and Rodrigo da Rosa Righi. 2020. Context awareness in healthcare: a systematic literature review. *Universal Access in the Information Society* 19, 2 (2020), 245–259. <https://doi.org/10.1007/s10209-019-00664-z>
- [8] Leonard J. Haas, J. G. Benedict and Joseph C. Kobos. 1996. Psychotherapy by telephone: Risks and benefits for psychologists and consumers. *Professional Psychology: Research and Practice* 27, 2 (1996), 154–160. <https://doi.org/10.1037/0735-7028.27.2.154>
- [9] Yunjo Han, Hyemin Lee, Kobiljon E. Toshnazarov, Youngtae Noh, and Uichin Lee. 2023. StressBal: Personalized Just-in-time Stress Intervention with Wearable and Phone Sensing. In *Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers* (Cambridge, United Kingdom) (*UbiComp/ISWC '22 Adjunct*). Association for Computing Machinery, New York, NY, USA, 41–43. <https://doi.org/10.1145/3544793.3560324>
- [10] Susan Holtzman, Drew DeClerck, Kara Turcotte, Diana Lisi, and Michael Woodworth. 2017. Emotional support during times of stress: Can text messaging compete with in-person interactions? *Computers in Human Behavior* 71 (2017), 130–139. <https://doi.org/10.1016/j.chb.2017.01.043>
- [11] H. Erin Lee and Jaehye Cho. 2019. Social Media Use and Well-Being in People with Physical Disabilities: Influence of SNS and Online Community Uses on Social Support, Depression, and Psychological Disposition. *Health Communication* 34, 9 (2019), 1043–1052. <https://doi.org/10.1080/10410236.2018.1455138> arXiv:https://doi.org/10.1080/10410236.2018.1455138
- [12] Siyue Li and Guanjin Zhang. 2021. Supportive communication on social networking sites: The impact of post valence and relational closeness on support provision. *Telematics and Informatics* 63 (2021), 101657. <https://doi.org/10.1016/j.tele.2021.101657>
- [13] Xiaolin Lin, Dawei Zhang and Yibai Li. 2016. Delineating the dimensions of social support on social networking sites and their effects: A comparative model. *Computers in Human Behavior* 58 (2016), 421–430. <https://doi.org/10.1016/j.chb.2016.01.017>
- [14] J. A. Naslund, K. A. Aschbrenner, L. A. Marsch and S. J. Bartels. 2016. The future of mental health care: peer-to-peer support and social media. *Epidemiology and Psychiatric Sciences* 25, 2 (2016), 113–122. <https://doi.org/10.1017/S2045796015001067>
- [15] National Institute of Mental Health. 2023. Major Depression. <https://www.nimh.nih.gov/health/statistics/major-depression> Accessed: 2024-06-05.
- [16] Hyun Jung Oh, Elif Ozkaya and Robert LaRose. 2014. How does online social networking enhance life satisfaction? The relationships among online supportive interaction, affect, perceived social support, sense of community, and life satisfaction. *Computers in Human Behavior* 30 (2014), 69–78. <https://doi.org/10.1016/j.chb.2013.07.053>
- [17] Adrian A. Ong, M. Boyd Gillespie and Jing Li. 2016. Overview of smartphone applications for sleep analysis. *World Journal of Otorhinolaryngology- Head and Neck Surgery* 02, 01 (2016), 45–49. <https://doi.org/10.1016/j.wjorl.2016.02.001>
- [18] World Health Organization. 2023. Depression. <https://www.who.int/news-room/fact-sheets/detail/depression> Accessed: 2024-06-05.
- [19] Mary E. Procidano and Kenneth. Heller. 1983. Measures of perceived social support from friends and from family: three validation studies. *Am J Community Psychol* 11, 1 (1983), 1–24. <https://doi.org/10.1007/bf00898416>
- [20] Stephen A Rains, Emily B Peterson and Kevin B Wright. 2015. Communicating social support in computer-mediated contexts: A meta-analytic review of content analyses examining support messages shared online among individuals coping with illness. *Communication Monographs* 82, 4 (2015), 403–430.
- [21] René Riedl. 2022. On the stress potential of videoconferencing: definition and root causes of Zoom fatigue. *Electronic Markets* 32, 1 (2022), 153–177. <https://doi.org/10.1007/s12525-021-00501-3>
- [22] Bobby Rozzell, Cameron W. Piercy, Caleb T. Carr, Shawn King, Brianna L. Lane, Michael Tornes, Amy Janan Johnson, and Kevin B. Wright. 2014. Notification pending: Online social support from close and nonclose relational ties via Facebook. *Computers in Human Behavior* 38 (2014), 272–280. <https://doi.org/10.1016/j.chb.2014.06.006>
- [23] Chris Salisbury, Alicia O’Cathain, Louisa Edwards, Clare Thomas, Daisy Gaunt, Sandra Hollinghurst, Jon Nicholl, Shirley Large, Lucy Yardley, Glyn Lewis, Alexis Foster, Katy Garner, Kimberley Horspool, Mei-See Man, Anne Rogers, Catherine Pope, Padraig Dixon, and Alan A. Montgomery. 2016. Effectiveness of an integrated telehealth service for patients with depression: a pragmatic randomised controlled trial of a complex intervention. *Lancet Psychiatry* 3, 6 (2016), 515–25. [https://doi.org/10.1016/s2215-0366\(16\)00083-3](https://doi.org/10.1016/s2215-0366(16)00083-3)
- [24] Karla Santo, Anna Singleton, Clara K Chow and Julie Redfern. 2019. Evaluating Reach, Acceptability, Utility, and Engagement with An App-Based Intervention to Improve Medication Adherence in Patients with Coronary Heart Disease in the MedApp-CHD Study: A Mixed-Methods Evaluation. *Medical Sciences* 7, 6 (2019). <https://doi.org/10.3390/medsci7060068>
- [25] Stefan Schneider, Doerte U Junghaenel, Tania Gutsche, Hio Wa Mak, and Arthur A. Stone. 2020. Comparability of Emotion Dynamics Derived From Ecological Momentary Assessments, Daily Diaries, and the Day Reconstruction Method: Observational Study. *J Med Internet Res* 22, 9 (2020), e19201. <https://doi.org/10.2196/19201>
- [26] Jay H. Shore. 2013. Telepsychiatry: videoconferencing in the delivery of psychiatric care. *Am J Psychiatry* 170, 3 (2013), 256–62. <https://doi.org/10.1176/appi.ajp.2012.12081064>
- [27] Jamie Travis, Kathryn Roeder, Heather Walters, John Piette, Michele Heisler, Dara Ganoczy, Marcia Valenstein, and Paul Pfeiffer. 2010. Telephone-based mutual peer support for depression: a pilot study. *Chronic Illness* 6, 3 (2010), 183–191. <https://doi.org/10.1177/1742395310369570> arXiv:https://doi.org/10.1177/1742395310369570 PMID: 20634226.
- [28] Kiona K Weisel, Lukas M Fuhrmann, Matthias Berking, Harald Baumeister, Pim Cuijpers, and David D Ebert. 2019. Standalone smartphone apps for mental health—a systematic review and meta-analysis. *NPJ digital medicine* 2, 1 (2019), 118.
- [29] Jennifer L. Welbourne, Anita L. Blanchard and Marla D. Boughton. 2009. Supportive communication, sense of virtual community and health outcomes in online infertility groups. In *Proceedings of the Fourth International Conference on Communities and Technologies* (University Park, PA, USA) (*CT '09*). Association for Computing Machinery, New York, NY, USA, 31–40. <https://doi.org/10.1145/1556460.1556466>
- [30] Peter Yellowlees, Michelle Burke Parish, Álvaro González, Steven Chan, Don Hilty, Ana-Maria Iosif, Robert McCarron, Alberto Odor, Lorin Scher, Andrés Sciolia, Jay Shore, and Glen. Xiong. 2018. Asynchronous Telepsychiatry: A Component of Stepped Integrated Care. *Telemed J E Health* 24, 5 (2018), 375–378. <https://doi.org/10.1089/tmj.2017.0103>